Table of contents

Electromagnectic Energy Exchange in Vacuum

charge conservation:

$$\frac{\partial}{\partial t}\varrho + \nabla \cdot j = 0 \tag{1}$$

$\varrho = \sigma\delta = \varepsilon_0 \nabla \cdot \mathcal{E} =$ charge density in interior

$\sigma =$ charge density at border (skin)

$\delta =$ Dirac delta-function

$j = \sigma\left(\mathcal{E} + v \times \mathcal{B}\right) = \nabla \times \mathcal{H} - \frac{\partial}{\partial t}\mathcal{D} = \frac{1}{\mu_0}\nabla \times \mathcal{B} - \varepsilon_0 \frac{\partial}{\partial t}\mathcal{E} \mathrel{\hat{=}}$ elecectric current density

$j \cdot \mathcal{E} \mathrel{\hat{=}}$ energy gain by fields $\mathcal{E}$ and $\mathcal{B}$ through current density j

$\mathcal{S} = \mathcal{E} \times \mathcal{H} \mathrel{\hat{=}}$ energy transport density (Intensity $= \left|\overline{\mathcal{S}}\right|$)

Energy conservation for (large extended) conductive medium:

$$\int \left(\frac{\partial}{\partial t}\left(w_e + w_m\right) + \ \cdot \mathcal{S} + j \cdot \mathcal{E}\right) dr = \frac{d}{dt}W_e + \frac{d}{dt}W_m + D_S + \int \left(j \cdot \mathcal{E}\right) dr \overset{\text{low frequency}}{=} \frac{1}{2}\frac{d}{dt}\left(LI^2\right) + \frac{1}{2}\frac{d}{dt}\left(\frac{Q^2}{C}\right) \tag{2}$$

Energy Balance

Electro-Magnetic Energy:

$$E = E_{kin} + E_{pot} = \sqrt{(pc)^2 + (mc^2)^2} \overset{\text{vacuum}}{=} pc = \hbar\omega \quad = \text{chargeenergy} + \text{fieldenergy} + \text{transportenergy} \tag{3}$$

Charge Energy:

$$\underbrace{j \cdot \mathcal{E}}_{\text{charge energy}} = -\left(\underbrace{\frac{\partial}{\partial t} \sum w_i}_{\text{field energy}} + \underbrace{\nabla \cdot \mathcal{S}}_{\text{transfer energy}} \right) = \mathcal{E} \cdot \nabla \times \mathcal{A} - \mathcal{E} \cdot \frac{\partial}{\partial t} \mathcal{D} \tag{4}$$

Field Energy:

$$\frac{\partial}{\partial t} w = \frac{\partial}{\partial t} \sum w_i = \frac{\partial}{\partial t} (w_e + w_m) = \frac{\partial}{\partial t} \left(\frac{\varepsilon_0}{2} \mathcal{E}^2 + \frac{1}{2\mu_0} \mathcal{B}^2 \right) \overset{vacuum}{=} \frac{\partial}{\partial t} \left(\frac{1}{2} \frac{1}{\mu_0} (\mathcal{A} \cdot \nabla \times \mathcal{B} - \mathcal{B}^2) \right) \tag{5}$$

Transfer Energy:

$$-\dot{w} \,\hat{=}\, \nabla \cdot \mathcal{S} \overset{matter}{=} \nabla \cdot (\mathcal{E} \times \mathcal{H}) \overset{vacuum}{=} \nabla \cdot (\mathcal{E} \times \mathcal{B}) \overset{\text{alternativ in vacuum}}{=} \nabla \cdot \left(\frac{1}{2} \frac{1}{\mu_0} \frac{\partial}{\partial t} (\mathcal{A} \times \mathcal{B}) \right) = \ldots \tag{6}$$

Local Energy Conservation:

Force $F = q (\mathcal{E} + v \times \mathcal{B})$

Work $W = \Delta E = Fr = pv = const.$

Power $\frac{d}{dt} W = 0$

$$\overbrace{\underbrace{j \cdot \mathcal{E}}_{\text{charge energy}} + \underbrace{\frac{\partial}{\partial t} \sum w_i}_{\text{field energy}}}^{=\text{ const.}} + \underbrace{\cdot \sum \mathcal{S}_{w_i}}_{\text{transfer energy}} \overset{alternatives}{=} \quad j \cdot \mathcal{E} + \frac{\partial}{\partial t} \left(\frac{1}{2} \varepsilon_0 \mathcal{E}^2 + \frac{1}{2} \frac{1}{\mu_0} \mathcal{A} \cdot \nabla \times \mathcal{B} \right) + \frac{1}{\mu_0} \nabla \cdot \left(\mathcal{E} \times \mathcal{B} + \frac{1}{2} \frac{\partial}{\partial t} \right. \tag{7}$$

conditions for modeling alternatives (Eigenvalue solution):

- $w = \sum w_i : \frac{\partial}{\partial t} w = -\nabla \cdot \mathcal{S}_w$
- $w = w(\mathcal{E}, \mathcal{B}) : \mathcal{E}\lambda^2 = \varrho\lambda$
- $w = w(\mathcal{E}, \mathcal{B}) : \mathcal{B}\lambda^2 = j\lambda$
- $\mathcal{S}_w = \mathcal{S}_w(\mathcal{E}, \mathcal{B}) : \mathcal{E}\lambda^2 = \varrho\lambda$
- $\mathcal{S}_w = \mathcal{S}_w(\mathcal{E}, \mathcal{B}) : \mathcal{B}\lambda^2 = j\lambda$

Energy Density Flux: Pyointing-Vector

$$\mathcal{S} \overset{vacuum}{=} \mathcal{E} \times \mathcal{B} \overset{matter}{=} \mathcal{E} \times \mathcal{H}$$

Far Field: Power $= \frac{\Delta E}{t} \hat{=} \frac{d}{dt} E = 0 \Rightarrow$ Energy $=$ const. $\Rightarrow$ Energy Conservation

$$\frac{d}{dt}E \hat{=} \frac{d}{dt}\left(\underbrace{\underbrace{\int_{\mathbb{R}^3} j \cdot \mathcal{E}}_{\text{charge energy}} + \underbrace{\int_{\mathbb{R}^3} \frac{\partial}{\partial t} \sum w_i}_{\text{field energy}}}_{= \text{const.}} \right) = 0$$

Near Field (local): Power $=$ Energytransport

$$\frac{d}{dt}E \hat{=} \frac{d}{dt}\left(\underbrace{\int_{\mathbb{R}^3} j \cdot \mathcal{E}}_{\text{charge energy}} + \underbrace{\int_{\mathbb{R}^3} \frac{\partial}{\partial t} \sum w_i}_{\text{field energy}} \right) \overset{local}{=} \underbrace{-\int_{\mathbb{R}^2} \mathcal{S} \cdot \mathbf{n} \underbrace{df}_{\text{"skin"}}}_{\text{transported energy density}}$$

Local Force Balance

Electro-Magnetic Force:

$$\dot{p} = F = q\left(\mathcal{E} + v \times \mathcal{B}\right) \quad = \text{chargeforce} + \text{fieldforce} + \text{stresstension} \tag{8}$$

Charge Force:

$$... = \varrho\mathcal{E} + j \times \mathcal{B} \tag{9}$$

Field Force:

$$... = \frac{\partial}{\partial t}\frac{1}{c^2}\mathcal{S} \tag{10}$$

Tension Force:

$$\nabla \cdot \mathcal{T} = \varepsilon_0\left(\mathcal{E} \times \nabla \times \mathcal{E} - \mathcal{E}\nabla \cdot \mathcal{E}\right) + \frac{1}{\mu_0}\left(\mathcal{B}\nabla \times \mathcal{B} - \mathcal{B}\nabla \cdot \mathcal{B}\right) \tag{11}$$

Local Momentum Conservation:

Force $F = q\left(\mathcal{E} + v \times \mathcal{B}\right) = \frac{d}{dt}p = 0$

Momentum $p = \int F = \frac{E}{c} = const.$

$$\overbrace{\underbrace{\varrho\mathcal{E} + j \times \mathcal{B}}_{\text{charge force}} + \underbrace{\frac{\partial}{\partial t}\frac{1}{c^2}}_{\text{field force}}\mathcal{S}}^{...} + \underbrace{\nabla \cdot \mathcal{T}}_{\text{tension force}} = 0 \tag{12}$$

Tension Force: Stress-Tensor

Stress Tensor quantifies tension $\hat{=}\Delta$ Forces = Pressure - Traction $\hat{=}$ "torsion tension" at angular momentum

$$\mathcal{T} \hat{=} \mathcal{T}_{ik} = \delta_{ik} \left(\frac{1}{2} \varepsilon_0 \mathcal{E}^2 + \frac{1}{2} \frac{1}{\mu_0} \mathcal{B}^2 \right) - \varepsilon_0 \mathcal{E}_i \mathcal{E}_k - \frac{1}{\mu_0} \mathcal{B}_i \mathcal{B}_k \tag{13}$$

$\mathcal{E}$ and $\mathcal{B}$ fields of spacial confined charge and current density distribution, far field, static: $\mathcal{T} \sim \frac{1}{r^4} \Leftrightarrow \int_{\mathbb{R}^3} \mathcal{T} d^3 r = \int_\infty \mathcal{T} df = 0$

Far Field: Force = 0 $\Leftrightarrow$ Momentum = const. $\Rightarrow$ Momentum Conservation

$$F = \dot{p} \hat{=} \frac{d}{dt} \underbrace{\left[\underbrace{\int_{local} (\varrho \mathcal{E} + j \times \mathcal{B})}_{\text{charge momentum}} + \underbrace{\frac{1}{c^2}}_{\text{field momentum}} \mathcal{S} \right]}_{=p=const.} = 0$$

Near Field (local): Force = Tension

$$F = \dot{p} \hat{=} \frac{d}{dt} \underbrace{\left[\underbrace{\int_{local} (\varrho \mathcal{E} + j \times \mathcal{B})}_{\text{charge momentum}} + \underbrace{\frac{1}{c^2}}_{\text{field momentum}} \mathcal{S} \right]}_{=p \neq const.,\ \exists -\mathcal{T} \text{ local tension}} \overset{local}{=} \underbrace{- \int_{\mathbb{R}^2} \mathcal{T} \cdot \mathbf{n} \underbrace{df}_{\text{"skin"}}}_{\text{stress tension appears}}$$

Momentum Balance

Momentum Conservation and Local Tension: linear momentum conservation if $p = const.$, otherwise "tension" occures:

$$p = \int F = \int \frac{d}{dt} p = \sqrt{\frac{1}{c^2} \left(E^2 - (mc)^2 \right)} \overset{\text{vacuum}}{=} \frac{E}{c} = \frac{2\pi}{c^2} h\nu$$

$p = const.$, linear, vacuum, Far Field:

$$F = \dot{p} \mathrel{\hat{=}} \int\limits_{\mathbb{R}^3} \frac{d}{dt} \left[\underbrace{(\varrho \mathcal{E} + j \times \mathcal{B})}_{\text{charge force density}} + \underbrace{\frac{\partial}{\partial t} \frac{1}{c^2}}_{\text{field force density}} \mathcal{S} \right] = \frac{d}{dt} \underbrace{\left[\underbrace{\int\limits_{\mathbb{R}^3} (\varrho \mathcal{E} + j \times \mathcal{B}) +}_{\text{charge momentum}} \underbrace{\frac{1}{c^2}}_{\text{field momentum}} \mathcal{S} \right]}_{= \text{ p momentum conserved if p = const.}} = 0 \tag{14}$$

$p \neq const.$, local $\in \mathbb{R}^2 \ll \mathbb{R}^3 = \infty$, linear, vacuum, Near Field $\Leftrightarrow \exists -\mathcal{T}$ as tension compensation flux:

$$F = \dot{p} \mathrel{\hat{=}} \int\limits_{local} \frac{d}{dt} \left[\underbrace{(\varrho \mathcal{E} + j \times \mathcal{B})}_{\text{charge force density}} + \underbrace{\frac{\partial}{\partial t} \frac{1}{c^2}}_{\text{field force density}} \mathcal{S} \right] = \frac{d}{dt} \underbrace{\left[\underbrace{\int\limits_{local} (\varrho \mathcal{E} + j \times \mathcal{B}) +}_{\text{charge momentum}} \underbrace{\frac{1}{c^2}}_{\text{field momentum}} \mathcal{S} \right]}_{= p \neq const., \, \exists -\mathcal{T} \text{ local tension}} \overset{local}{=} - \int\limits_{\mathbb{R}^2} \mathcal{T} \cdot \mathbf{n} \, \underbrace{d\!f}_{\text{"skin}} \tag{15}$$

Momentum $p = \int\limits_{\mathbb{R}^3} (\varrho \mathcal{E} + j \times \mathcal{B}) + \frac{1}{c^2} \mathcal{S}$

photon Spin $= \int\limits_{\mathbb{R}^3} \frac{1}{c^2} \mathcal{S}$

angular momentum (photon Spin):

$$D = \dot{L} \mathrel{\hat{=}} \dots \tag{16}$$

Radiation Pressure

average energy density $\hat{=}$ radiation pressure, if stationary (very slow flow motion $v \ll c$) :

$$\overline{w} \mathrel{\hat{=}} -\nabla \mathcal{T} \cdot \mathbf{n} \mathrel{\hat{=}} -\frac{1}{c} \left| \overline{\mathcal{E} \times \mathcal{H}} \right| = -\frac{1}{c} \left| \overline{\mathcal{S}} \right| = \overline{(\varrho \mathcal{E} + j \times \mathcal{B}) + \frac{\partial}{\partial t} \frac{1}{c^2} \mathcal{S}} \overset{stationary}{=} \overline{\varrho \mathcal{E} + j \times \mathcal{B}} \tag{17}$$

Abbreviations:

w_i = energy densities (electric w_e, magnetic e_m, gravitational, etc. ...), $i \in \mathbb{N}$

$\mathcal{S}_{w_i}$ = Poynting-Vectros $\hat{=}$ energy transport densities (transfers) by fields (electric, magnetic, gravitational, etc. ...) to corresponding energy densities w_i

$j \cdot \mathcal{E}$ = energy gain provided by fields $\mathcal{E}$ and $\mathcal{B}$ through current density j

$\mathcal{T}$ = Stress Tensor, (tension, momentum flow density)

$\mathbf{n}$ = surface normal vector (orthongoal on skin)

$\Delta\cdot :=$ div grad = Laplacian

J = "ignition of inhomogenity,cause"

j = Current density

ϱ = Charge density

V = Potentialdifference

I = Current

R = Resistance

L = Inductance

C = Capacitance

Q = charge (ion)

$\Psi = LI$ = magn. Flux

$\mathcal{S} = \mathcal{E} \times \mathcal{H}$ = Poynting-Vector, radiation intensity

$D_S = \int (\,\cdot\, \mathcal{S})\, dr = \int \mathrm{div}(\mathcal{E} \times \mathcal{H})\, dr$ = dissipation engergy lost from radiation ≈ 0 for low frequencies ω

$d = \sqrt{\frac{1}{\mu\sigma}\frac{2}{\omega}}$ = skin depth (surface) of conductor

$r = \frac{d}{\sqrt{2i}}\rho = \frac{d}{1+i}\rho$ = radius (extension) of conductor

$\rho = kr$ = "radius of curvature k"

μ = permeability of conductor

σ = conductivity of conductor (dielectricum)